Huawei Access Node BAS-ACC-73 Routing Protocol Flap Resolution SOP

Node: BAS-ACC-73 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_8187

Description

The Huawei Access Node BAS-ACC-73 routing protocol flap failure is a complex issue that involves the dynamic interaction between the Border Gateway Protocol (BGP), Open Shortest Path First (OSPF), and Intermediate System to Intermediate System (ISIS) routing protocols. When the BGP route churn, OSPF adjacency loss, or ISIS LSP flooding occurs on this specific role and vendor equipment, it can lead to a situation where the routing protocols are unable to stabilize, resulting in a routing protocol flap. This situation can be exacerbated by various factors such as misconfiguration, hardware failures, or software bugs. The Huawei Access Node BAS-ACC-73 is equipped with the Huawei NE5000E series router, which is a high-performance, high-capacity router that supports multiple routing protocols, including BGP, OSPF, and ISIS. The routing protocol flap failure on this node can have significant impacts on the overall network performance, availability, and reliability. Therefore, it is essential to identify the root cause of the issue and take corrective actions to resolve the problem.

Pre-requisite Checks

- check: Verify that the node BAS-ACC-73 is powered on and the Huawei NE5000E series router is functioning properly.

- check: Check the redundant link status to ensure that it is working correctly.

- check: Confirm that the BGP, OSPF, and ISIS routing protocols are enabled and configured correctly.

Potential Root Causes

Cause	Probability
Misconfigured BGP routing policy	High
Hardware failure in the NE5000E series router	Medium
Software bug in the Huawei firmware	Low

Diagnostic Commands & Expected Outputs

```
Command: display bgp routing-policy, Expected Output: The output will display the BGP routing policy configuration., Description: This command is used to diagnose the misconfigured BGP routing policy issue by displaying the BGP routing policy configuration.

Command: display ospf routing-table, Expected Output: The output will display the OSPF routing table., Description: This command is used to diagnose the OSPF adjacency loss issue by displaying the OSPF routing table.
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```
Command: display isis database, Expected Output: The output will display the ISIS database., Description: This command is used to diagnose the ISIS LSP flooding issue by displaying the ISIS database.
```

Resolution Steps

Step 1: Step: Clear the BGP routing policy configuration, Command: reset bgp routing-policy, Description: This step involves clearing the BGP routing policy configuration to resolve the misconfigured BGP routing policy issue.

Step 2: Step: Replace the faulty NE5000E series router hardware, Command: replace hw NE5000E series router, Description: This step involves replacing the faulty NE5000E series router hardware to resolve the hardware failure issue.

Step 3: Step: Upgrade the Huawei firmware, Command: upgrade firmware, Description: This step involves upgrading the Huawei firmware to resolve the software bug issue.

Post-Check Verification

```
Command: display bgp routing-policy, Expected Output: The output will display the BGP routing policy configuration., Description: This command is used to verify that the BGP routing policy configuration has been resolved.

Command: display ospf routing-table, Expected Output: The output will display the OSPF routing table., Description: This command is used to verify that the OSPF adjacency loss issue has been resolved.

Command: display isis database, Expected Output: The output will display the ISIS database., Description: This command is used to verify that the ISIS LSP flooding issue has been resolved.
```

Roll-back Procedures

- Step: Restore the previous BGP routing policy configuration, Command: restore bgp routing-policy, Description: This step involves restoring the previous BGP routing policy configuration in case the fix fails.

- Step: Remove the new NE5000E series router hardware, Command: remove hw NE5000E series router, Description: This step involves removing the new NE5000E series router hardware in case the fix fails.

- Step: Downgrade the Huawei firmware, Command: downgrade firmware, Description: This step involves downgrading the Huawei firmware in case the fix fails.